

JavaScript Objects & Destructuring

Creating Objects

```
const student = {  
  name: 'Ahmed Khan',  
  age: 20,  
  marks: 85,  
  city: 'Lahore'  
};
```

Accessing Properties

```
student.name; // Ahmed Khan  
student['age']; // 20
```

What is Destructuring?

Destructuring extracts properties from objects into separate variables (MUCH CLEANER!).

Before Destructuring:

```
const name = student.name;  
const age = student.age;  
const marks = student.marks;
```

After Destructuring (Much Better!):

```
const { name, age, marks } = student;
```

Destructuring with Renaming:

```
const { name: studentName, age: studentAge } = student;  
console.log(studentName); // Ahmed Khan
```

Nested Destructuring:

```
const user = {  
  profile: {
```

```
name: 'Ali',  
contact: { email: 'ali@gmail.com' }  
}  
};  
  
const { profile: { name, contact: { email } } } = user;
```

In Function Parameters (Super Clean!):

```
function displayStudent({ name, age, marks }) {  
  console.log(name, age, marks);  
}  
  
displayStudent(student);
```

With Array Methods:

```
const students = [  
  { name: 'Ahmed', marks: 85 },  
  { name: 'Umer', marks: 92 }  
];  
  
// Get names using map + destructuring  
const names = students.map(({ name }) => name);
```